# Rakshith Suresh

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MS EE student who spent 10 months building a RISC-V matrix-multiply SoC from RTL to GDS on Sky130, ASAP7 and IHP, now signed off on IHP SG13CMOS5L with the tapeout handled by Symbiotic EDA. Seeking Spring 2027 co-op and full-time hardware roles from May 2027.

## Education

**University of Southern California** — Los Angeles, CA
*Master of Science in Electrical Engineering, VLSI and Digital Design* — *Aug 2025 – May 2027*
- Coursework: MOS VLSI Design (EE477), VLSI System Design (EE577a, EE577b), Computer Systems Organization (EE457)

**Visvesvaraya Technological University** — Bangalore, India
*Bachelor of Engineering in Electronics and Communication Engineering* — *Jan 2020 – May 2024*

## Technical Skills

**Verification:** UVM, SVA, Formal Verification (SymbiYosys/Z3), Functional Coverage, Gate-Level Simulation, VCS, Verilator, cocotb, uvm-python

**HDL & Programming:** SystemVerilog, Verilog, Python, C, C++, TCL, Perl

**Physical Design:** Floorplanning, Timing Closure, IR Drop, DRC/LVS, Cadence Innovus, Synopsys PrimeTime, OpenROAD, LibreLane, KLayout

**Process Technologies & Tools:** SkyWater 130nm, IHP SG13G2/SG13CMOS5L 130nm, GF 180nm, ASAP7 7nm, FPGA (Vivado, Artix-7)

## Projects

**Argo: RISC-V + Systolic Array SoC on IHP 130nm [GitHub] [Write-up]** — Jan 2026 – Present
*SystemVerilog, LibreLane, OpenROAD, Yosys, KLayout, Icarus – RTL to GDS on Sky130, ASAP7, IHP SG13G2/SG13CMOS5L*
- **3 process nodes**: Submitted the systolic array to the ChipFoundry CI2609 Sky130 shuttle, hardened the SoC on ASAP7 (500 MHz, 0 DRC), and signed off Argo on IHP SG13CMOS5L.
- **Argo SoC, 100 MHz typical, QFN32**: PicoRV32 SoC with an 8x8 int8 systolic array, 2x4 KB IHP SRAMs, SPI boot, UART and a detachable 2 KB SRAM block for silicon validation on its own supply; 2.2x2.2 mm die, hold met at all corners.
- **Signoff-clean GDS with KLayout only**: 0 DRC, density and antenna errors, LVS match and 36 mV worst IR drop, using local LVS, fill and bond-pad fixes for a PDK with no Magic support; built and checked on Symbiotic EDA's BlueGarden CI/CD.
- **Slow-corner slew violations cut 95% (1,297 to 66)**: Traced to up to 200 antenna diodes per bit on the CPU write-data bus; capping repair wire length buffered the long nets and improved slow-corner WNS from $-2.07$ to $-1.38$ ns.
- **Gate-level POST passing in zero-delay and SDF**: Root-caused a gate-level multiply hang to X-pessimism in a non-reset PicoRV32 counter, proved it with a netlist probe, and passed a power-on self-test of every memory on the routed netlist.

**Custom ISA Hardware Accelerator [GitHub]** — Jun 2026 – Present
*SystemVerilog, Python, C, Verilator, Vivado – RISC-V processor with a custom hardware accelerator*
- **27/27 tests passing**: Extended a RISC-V (RV32I) processor with 3 custom matrix-multiply instructions, validating every instruction against a Spike ISS reference model.
- **757 MHz, 0 DRC on ASAP7 7nm**: Closed setup and hold timing across 10,930 cells, then brought the design up on an Arty A7 FPGA, independently root-causing and resolving a board-level reset polarity issue during bring-up.

**UVM Verification Environment for 8x8 Systolic Array [GitHub]** — Apr 2026 – Present
*uvm-python, cocotb, Python, Verilator, VCS – Self-checking UVM testbench for a matrix-multiply accelerator*
- **1,337/1,337 checks passed, 0 mismatches**: Built a self-checking UVM environment (sequencer, driver, monitor, scoreboard) with a coverage plan closed across 5 regression seeds, using an independent reference model.
- Root-caused a VCS vs. Verilator discrepancy to a timescale LRM violation, restoring identical results across both tools.

**Multi-Warp SIMT GPU Core in RV32I [GitHub]** — Jun 2026 – Present
*SystemVerilog, Yosys, OpenROAD, ASAP7 – 2-warp, 8-lane SIMT execution engine*
- **34/34 tests passing, 500 MHz, 0 DRC**: Designed a round-robin-scheduled, 2-warp SIMT core (283 cells plus 8 macros, 2,863 $\mu m^2$, ASAP7 FF corner, +1,454 ps setup and +50 ps hold slack) and verified both threads against Spike ISS.

**RV32I 5-Stage Pipeline CPU [GitHub]** — May 2026
*SystemVerilog, SymbiYosys, Z3 – Branch predictor, hazard unit, formal verification*
- **27/27 tests, 5 formal properties**: Proved 5 SVA properties on the hazard unit with SymbiYosys and co-simulated against Spike ISS.

**Custom Development Board** — In Progress
*PCB design – teaching myself board-level design to bring up my own chip*

## Experience

**Toyota Kirloskar Motor Private Limited** — Bangalore, India
*Electrical Engineering Intern* — *Aug 2023 – Oct 2023*
- Diagnosed and resolved electrical faults across 12 production line systems by analyzing schematics and identifying root cause.

## Leadership

- Viterbi Graduate Mentor, USC. Member, IEEE USC Student Branch. Class Representative, VTU (4 years). Project Lead, inter-state pitch competition, 3rd place finish.